Auditable reasoning record

Reciprocal-simplex agentic research rerun

> Important distinction. This document does not contain raw private chain-of-thought. Those internal reasoning tokens are not exposed. It compiles the structured reasoning summaries, decision records, verification checks, and usage metadata that were explicitly saved as auditable project artifacts.

Run ID: `20261019T114252Z_cox_rerun`

Compiled: 20 October 2026

`Official capability reference: https://developers.openai.com/api/docs/guides/reasoning`

Contents

1. Run provenance and scope

2. Literature agent outcome

3–7. Proof-agent reasoning summaries (Poisson cluster, Cox, Hawkes, selected construction, necessary conditions)

8–12. Critical-review reasoning summaries and decision records

13. Final synthesis reasoning summary and decision record

14. Recovered token and runtime ledger

15. Source inventory and SHA-256 provenance

1. Run provenance and scope

The frozen run manifest records model gpt-5.6-sol with reasoning effort xhigh, no fallback, fresh contexts, and a completed workflow. The governing exact target is recorded as unresolved.

Field	Recorded value
Run interval	2026-10-19T11:42:52Z — 2026-10-19T15:19:35Z
Model / effort	gpt-5.6-sol / xhigh
Target	exact full product densities g2, g3, g4; prescribed intensity
Strongest positive result	Exact full g2, g3 and g4 identities on the complete domains, with no positive pair cutoff, for every prescribed intensity in a nonempty geometry-dependent interval starting at zero.
High-intensity status	Neither general existence nor general non-existence established.

Interpretation of the record

A reasoning-token count measures hidden output-token usage; it is not a recoverable text transcript. The sections below are the durable, author-visible reasoning records deliberately required by the workflow.

2. Literature agent — outcome and scope

Source: runs/20261019T114252Z_cox_rerun/literature/dossier.md

Overall literature-check status: E. No source establishing the governing target was located in the supplied collection or in the bounded wider search described below. Here E means only no supporting result located within this documented search. It does not mean that the statement is novel, that no proof exists, or that the problem is known to be open.

The target checked is existence, for every prescribed admissible `(lambda_0,M,h_3,h_4)`, of one stationary simple locally finite point process in `R^3`, with intensity exactly `lambda_0`, finite local moments through order four and absolutely continuous factorial moment measures through order four, whose full normalized factorial/product densities obey

```
g_2=C_2/D,\qquad g_3=C_3/A,\qquad g_4=C_4/V
```

almost everywhere on the three finite physical domains in the frozen problem statement. In particular, the pair domain contains every `0<D<=M`; there is no positive lower distance cutoff. The triangle and tetrahedron domains impose the stated absolute altitude cutoffs. The constants `C_2,C_3,C_4` may depend on the prescribed parameters but must be positive. Approximation, connected densities, equality only for same-cluster terms, a selected intensity, or a strictly smaller domain does not settle this target.

The closest supplied results are both D (partial):

- the local spherical Cox construction gives uniform relative approximation on

the target-shaped finite domains at independently prescribed intensity, but not exact identities at any finite parameter choice;

- the local finite-cluster construction gives exact full identities, but has a

positive pair cutoff and chooses a small intensity together with its amplitudes. It is not presented as a Cox process.

The basic Cox moment identity, the Poisson-cluster partition formula and the Hawkes cumulant formulas are A for the structural statements they actually make; none is an existence theorem for the reciprocal-simplex target. Two restricted model classes can be excluded by short consequences of primary formulas: ordinary log-Gaussian Cox processes and Hermitian determinantal point processes are C class-specific nonmatches. These exclusions do not apply to general Cox processes or to general point processes.

3. Proof agent 1 — Poisson cluster process

Source: runs/20261019T114252Z_cox_rerun/tracks/approach_1/submissions/v02.md

1. Starting point. The exact finite-cluster manuscript already solves both triangular inverse problems on a domain with

\delta>0, but its intensity and amplitudes are selected together. The spherical carrier manuscript supplies prescribed intensity but only approximate identities.

2. Pair-cutoff test. Every place where \delta entered the finite-cluster proof was checked. The only potentially singular integral is

\int_{|y|\le M}|y|^{-1}dy, which equals 2\pi M^2. The positivity and marginal estimates are uniform down to D=0. Thus setting \delta to zero is legitimate; no limiting process is needed.

3. Altitude-order test. Directly using the target triangle support fails to be deletion-nested when h_4<h_3. Enlarging the auxiliary triangle support to altitude at least

\min(h_3,h_4) repairs nesting and strengthens, rather than weakens, the triangle identity.

4. Prescribed-intensity choice. Setting a_2=\lambda_0/I makes the exact bookkeeping intensity equal to the input

\lambda_0. The remaining positivity conditions become the explicit polynomials (6). Their common feasible set is the compact interval (7), so the upper endpoint is included rather than merely approached.

5. Alternatives rejected. Independent thinning preserves normalized g_n but can only lower intensity. Independent superposition changes every connected-to-full partition coefficient. Spatial rescaling preserves reciprocal form but also preserves the dimensionless obstruction scale

\lambda M^3; it cannot turn the present small-intensity theorem into an arbitrary-intensity theorem on the same physical domain.

6. Why the result stops. The pair law forces a same-parent pair mass of order

\lambda^2M^3. Within the class covered by Proposition 2, a cluster containing at most L points supplies at most L-1 same-parent partners per point, yielding (11). Therefore the present max-four refinement cannot meet the missing arbitrary-intensity quantifier. The argument does not decide whether a growing or unbounded cluster size, or a different process class, can do so.

4. Proof agent 2 — Cox processes

Source: runs/20261019T114252Z_cox_rerun/tracks/approach_2/submissions/v02.md

1. I formulated the Cox step in both directions. The forward direction lifts a stationary diffuse directing measure to a stationary simple point process with the required moments and densities. The converse uses uniqueness of the Cox directing law, the diagonal mass of \mathbb E\Lambda^{\otimes2}, and the fourth mixed-Poisson moment. This justifies calling the remaining question a directing-measure moment problem rather than only a sufficient ansatz.
2. I made the sphere-incidence step measure-theoretic. The nonnegative coarea identity applies to the measurable weights p(r)b(r)^j, the orthogonal fibre has dimension 4-j, and the normal-Jacobian ratio is r^{j-1}/((j-1)!\Delta_{j-1}). Local finiteness converts the extended-valued coarea expression into a locally integrable density. At order four the formula uses one fixed Borel version of p b^4, with version independence only asserted almost everywhere.
3. A finite-cutoff curvature argument would exclude the original sphere parameters but not the entire radial class. The explicit construction (5.7)--(5.8) confirms that radial mixing can repair the full pair law. The robust obstruction is instead that Aq_3 depends only on circumradius. Combining this with every order-three partition forces a shape-constant expression that the fixed-radius isosceles family disproves.
4. I tested two possible bridges beyond the sphere class. Finite-pointwise-second-moment intensity fields cannot create the required pair singularity, and the specific exact finite-cluster witness has a non-mixed-Poisson count distribution in one large positive-measure window. Neither argument covers arbitrary diffuse random measures.
5. Window moment positivity yields (8.1) and (8.2), but the prescribed identities leave lower-altitude third- and fourth-order configurations free. I therefore retain the general Cox problem as unresolved rather than filling the omitted configurations by an unrequested formula.

5. Proof agent 3 — Hawkes process

Source: runs/20261019T114252Z_cox_rerun/tracks/approach_3/submissions/v02.md

1. Spatialisation choice. A temporal Hawkes process alone does not define

hypocentres in \mathbb R^3. Projection over all time has infinite spatial intensity, while a deterministic slice is empty for a diffuse process. A stationary space--time process observed on a finite time interval gives a genuine stationary spatial process on all of \mathbb R^3.

2. Intensity choice. Subcriticality gives intensity \mu/(1-\eta).

Equation (8) meets the independently prescribed spatial intensity exactly without rescaling the physical geometric domains.

3. Moment check. The Borel total-progeny law has an exponential moment when

\eta<1. Equations (5)--(7) then give a direct Poisson-cluster proof of finite local fourth moments.

4. Measure-first hierarchy. Fixed-root clusters necessarily contain a root

Dirac mass. I therefore kept the recursion as a measure identity, proved that integrating over the diffuse immigrant location removes that mass, and only then used densities and the finite-time spatial pushforward.

5. Partition check. The fixed-root recursion supplies connected

factorial-cumulant measures. Applying every set partition afterwards gives exactly 2,5,15 full product-density terms at orders two, three and four.

6. Critical alternative discarded for this theorem. At \eta=1, the

Poisson-immigrant Galton--Watson cluster has infinite mean progeny, so (4) fails. Special critical no-ancestor models have different hypotheses and do not supply the required fourth moments or exact reciprocal-simplex densities.

7. Decisive obstruction and its limit. For

\phi\in L^1\cap L^\infty, the descendant resolvent and the projected pair correlation are bounded, contradicting the required 1/D divergence. An integrable unbounded kernel or a marked carrier can evade this step, so those cases remain unresolved.

6. Proof agent 4 — agent-selected finite-cluster construction

Source: runs/20261019T114252Z_cox_rerun/tracks/approach_4/submissions/v02.md

1. Selected route. The dependent finite-cluster inversion was selected because it is the only fully inspected local route whose existing proof already gives exact full densities through order four. The construction explicitly subtracts every lower partition and then inverts the marginal hierarchy to obtain genuine nonnegative cluster laws.
2. Zero-cutoff check. The previous lower pair cutoff was checked against the actual estimates. It is not needed for integrability: in three dimensions $1/D$ has finite anchored mass at zero, and the positivity and marginal estimates remain valid with $S_2=\{0<D\leq M\}$.
3. Support-nesting check. Using the target triangle threshold directly fails when $h_4<h_3$, because a target tetrahedron's faces need only have altitude at least h_4. Enlarging the auxiliary triangle support to threshold $\min\{h_3,h_4\}$ repairs this without shrinking any requested domain or changing the target identity.
4. Prescribed-intensity check and provenance. Setting $a_2=\lambda_0/l$ makes the final intensity identity exact. The residual positivity errors are the explicit polynomials (3), whose vanishing at zero yields (4). This quantifies, for the zero-cutoff supports, the small-prescribed-intensity interval already implicit in the prior polynomial-in-amplitude theorem; it is not an arbitrary-intensity transformation.
5. Reanchoring and full-density check. The unit-Jacobian map (14), together with symmetry and translation invariance, makes the anchored cluster law independent of the distinguished daughter and validates the coefficient in (19). The order-four calculation then includes all fifteen set partitions. The q_4 definition cancels six pair-singleton-singleton terms, three pair-pair terms, four triple-singleton terms and the all-singleton term.
6. Alternatives not promoted. The spherical Cox route preserves arbitrary prescribed intensity but remains approximate at finite parameters; no valid exact directing-measure limit was found. A Gibbs existence theorem would not identify its interaction weight with the full correlation hierarchy. The truncated-slab construction appears only as a summary in the inspected frozen notes, not as a complete source supplied in this packet, and its arbitrary prescribed-intensity quantifier was not established here.
7. Why the theorem stops. The positivity proof is a small-intensity argument. The reduced-measure estimate (O) shows that fixed bounded cluster size is itself incompatible with arbitrarily high intensity at fixed M. Claiming the unrestricted target would require a genuinely different intensity mechanism, such as an exact Cox directing measure or clusters whose size grows without a fixed cap.

7. Proof agent 5 — non-existence and necessary conditions

Source: runs/20261019T114252Z_cox_rerun/tracks/approach_5/submissions/v02.md

- The pair singularity is locally integrable in three dimensions. The tetrahedral flattening divergence is removed by the prescribed h_4>0, and large-separation occupancy arguments cannot be used because the identities stop at diameter M.
- A direct attempt to infer fourth-order cumulants from the triangle target fails: faces of a target tetrahedron are not required to belong to \mathcal D_3. Formula (24) keeps those values explicit.
- The finite-cluster source was tested at its excluded endpoint \delta=0. The only newly singular integral is (8), which is finite. Every marginal estimate, pointwise positivity estimate, anchored normalisation, simplicity argument and parent-partition identity survives.
- Setting h=\min(h_3,h_4) resolves the face-support issue without shrinking the requested triangle domain. Setting a_2=\lambda _0/I prescribes the intensity. The residual kernels stay nonnegative when the explicit polynomials (6) are at most 1/2, which holds on a nonempty interval of prescribed intensities.
- The restricted theorem rules out the claim that no intensity admits a solution. It does not decide existence or non-existence for any prescribed \lambda _0>\lambda _*. For arbitrary intensity, count positivity yields (28)–(30), but no contradiction because the reciprocal amplitudes are free.
- Determinantal and ordinary log-Gaussian Cox failures remain class-specific and were not promoted to statements about general point processes or general Cox processes.
- Reused formulas were traced to the exact manuscript sections and are rederived here. Prior audit documents were treated only as historical records, not as independent support.

8. Critical review 1 — Poisson cluster

Source: runs/20261019T114252Z_cox_rerun/reviews/approach_1/review_v02.md

The following checks determined the verdict.

1. Scope and quantifiers. I compared the claim-status paragraph, Theorem 1, the post-theorem scope statement, Proposition 2 and the final limitations with the governing quantifiers. Each now distinguishes the certified interval from arbitrary prescribed intensity. No sentence upgrades the restricted construction or the bounded-cluster proposition into a general solution or obstruction.
2. Endpoint feasibility. The feasible set is the intersection, in [0,\infty), of three closed sublevel sets of continuous increasing functions. It contains a positive neighbourhood of zero and is bounded because E_2(t)\to\infty. Monotonicity makes it an initial interval, so it is exactly [0,\lambda_*] and the supremum is attained. Rechecking the downstream inequalities at equality shows that non-strict feasibility is enough: (18) retains the factor 1/2, while (23)--(24) retain the factor 1/4. Thus the endpoint produces genuine nonnegative cluster densities and is not merely a limiting value.
3. Permutation and re-rooting. I re-derived the symmetry induction from (14)--(16) and (22). The one-coordinate re-rooting map is an invertible integer linear map with inverse given by the opposite re-rooting, so its determinant is \pm1; applying it to three coordinates preserves absolute Jacobian one. For an ordered r-tuple there are exactly (j)_r injective daughter assignments, and the parent-rate/mark-density product is F_j/j!. The resulting coefficient is exactly 1/(j-r)!, with no missing factor j, r! or j!.
4. Intensity bookkeeping. Integrating the second inversion gives Z_4=Q_4, Z_3=Q_3-Q_4 and Z_2=Q_2-Q_3+Q_4/2. Hence the non-singleton intensity is Q_2-Q_3/2+Q_4/6. The singleton rate in (28), together with Q_2=\lambda-\lambda^2J, raises this to exactly \lambda. The endpoint argument does not alter this calculation.
5. Full rather than connected densities. Equations (31)--(33) contain every set partition: 2 terms at order two, 5 at order three and 15 at order four. The recursive definitions cancel those full lower-partition terms on the deletion-nested supports. The proof never identifies a same-parent kernel with a full product density.
6. Domains and singularities. The pair identity is proved on all 0<D\le M, and the only new singular anchored integral is finite in three dimensions. The triangle support contains \mathcal D_3; every face required in the tetrahedral cancellation lies in that support. No formula outside the governing finite domains is used to infer the final identities.
7. Process properties. Bounded cluster size and offsets imply local finiteness and finite local exponential moments. Absolute continuity of the mark laws makes within-cluster collisions null; diffuse independent parents and marks exclude cross-cluster collisions. The block-density partition formula gives absolute continuity at every finite order.
8. Obstruction scope. The proof of Proposition 2 uses only nonnegativity of the same-parent pair density and (N)_2\le(L-1)N. The latter depends essentially on the fixed almost-sure bound N\le L, so the revised exclusions are mathematically necessary and correctly stated.

These checks are an auditable summary of the decisive mathematical tests, not a claim of machine formalisation.

Decision record

Decision	Evidence	Result
Accept the repaired scope statement	Claim status; Theorem 1 and its following paragraph; Proposition 2 final paragraph; `Checks and limitations`	Accepted as a restricted theorem and scoped obstruction only.
Include \lambda_*=\max\Lambda	Equations (6)--(7a); closedness, interval property, positive neighbourhood and boundedness; downstream non-strict positivity checks	Accepted; endpoint attainment and endpoint construction are valid.
Accept the ordered-daughter normalisation	Lemma 3; equation (29); explicit (j)_r assignment count against Z_j/j!	Accepted; coefficient 1/(j-r)! is correct.
Preserve the unchanged exact-density conclusion	Equations (14)--(24), (26)--(36); support nesting and complete partitions	Accepted on the certified intensity interval and stated nonvacuous geometry regime.
Require another repair round	No assigned issue remains and no new material defect was found	No.

9. Critical review 2 — Cox

Source: runs/20261019T114252Z_cox_rerun/reviews/approach_2/review_v02.md

The following decisive checks were performed independently on the frozen v02 formulas.

1. Cox converse. I checked each recovered property separately: uniqueness of the directing law from the restricted Laplace functional; diagonal mass

$M_2(\mathrm{diag}\cap B^2)=\mathbb E\sum_{x\in B}\Lambda\{x\}^2$; the exact fourth Touchard polynomial; and transfer of absolute continuity through $\alpha_X^{(n)}=M_n$. No converse condition is left asserted without a mechanism.

2. Incidence constants. For $j=2,3,4$, the fibre dimensions are $2,1,0$. Substituting these into the repaired coarea formula reproduces respectively

```
\frac{2\pi}{D}\int_{D/2}^L p(r)b(r)^2r^2\,dr, \quad
\frac1A\int_{R_3}^L\frac{p(r)b(r)^3r^3}{\sqrt{r^2-R_3^2}}\,dr, \quad
\frac{p(R_4)b(R_4)^4R_4^3}{6V},
```

with the last expression understood through the fixed Borel version. These checks confirm the factorial and polar factors in the general formula.

3. Local finiteness and a.e. status. The bound

$Q_j(K^j)\leq |K^{+L}|\kappa^j\int_0^L p(r)\,dr$ is sufficient for a locally integrable density. Equality of two almost-everywhere choices of $p b^4$ first gives the same measure and hence only an almost-everywhere equality of the resulting tuple-density versions, exactly as claimed.

4. Full hierarchy. Expanding the Poisson-carrier moments by labelled set partitions gives the displayed $2,5,15$ terms with no extra factorial. Substitution of the forced q_2 into the order-three expansion gives the constant $+2$ in equation (5.2); it is not a connected-density substitution.

5. Shape obstruction. I recomputed

$A=4R^2x(1-x^2)^{3/2}$ and $A(1/a+1/b+1/c)=R(1-x^2)(1+4x)$ for the fixed-R isosceles family. The endpoint contradiction is valid for every $C_2>0$. Positive measure of the triangle domain gives the strict interval needed around equilateral configurations.

6. Pair-only construction. Integrating the two pieces of f gives

$2\pi\int_{D/2}^L f=C_2-D$ for all $0<D\leq M$. The repaired requirement $L>M/2$ removes the zero-denominator problem when the positive tail is present. The definitions of a,b,p give $p b^2r^2=f$, intensity one, integrable p, and bounded carrier mass.

7. Window argument. Only one sufficiently large bounded positive-measure window is needed. Bounded offsets and positive type-four rate yield $0<\nu_4(B)<\infty$. The compound-Poisson pgf is entire, while a mixed-Poisson representation gives a holomorphic Laplace transform on the right half-plane. The identity theorem and the positive quartic leading term give the contradiction.

8. Global scope. None of these calculations constrains every diffuse directing measure, every irregular density field, dependent or non-spherical carriers, or the full fourth-order target. Section 8 correctly leaves densities on excluded low-altitude configurations free, so its moment-matrix discussion supplies necessary conditions only.

10. Critical review 3 — Hawkes

Source: runs/20261019T114252Z_cox_rerun/reviews/approach_3/review_v02.md

The following decisive checks were re-established from v02 rather than accepted from its response section.

1. Quantifier and target check. The governing target is an exact statement about full product densities for one process at every prescribed admissible $\lambda _0,M,h_3,h_4$. V02 does not claim to solve it. Its theorem quantifies only over the scalar linear subcritical $L^1\cap L^\infty$ Hawkes subclass and uses the full pair domain $0<D\le M$. The limitation appears in the theorem statement, remaining obligation, checks and response to review.
2. Fourth-moment check. The Borel coefficient asymptotics give radius $e^{\eta-1}/\eta>1$. The translation identity behind (5) is valid term by term, because each intersection of translates of W has measure at most $|W|$. This supplies integrable domination for differentiating the marked-Poisson exponential formula through order four. Equation (7) then gives the ordinary fourth moment. The argument also covers $\eta=0$ separately.
3. Measure-to-density check. In recursion (10), the root factor can be differentiated at most once, so a fixed-root term has at most one Dirac coordinate. Every $\mathsf H_B^u$ occurring in the source term has $B\subsetneq A$; integrating its possible root coordinate against the offspring density makes it absolutely continuous. The δ_0 part of the resolvent preserves the one fixed-root Dirac, while the Ψ-part smooths it. Immigrant integration removes the last Dirac. The local bound (16a) supplies local finiteness of the resulting Radon--Nikodym densities.
4. Partition check. Normalising (23)--(25) gives $g_3=1+\sum c_2+c_3$ and $g_4=1+\sum c_2+\sum c_2c_2+\sum c_3+c_4$. Substitution of the pair target reproduces exactly the constants and signs in (31), while rearrangement gives (32). The fourth-order formula contains all 15 set partitions.
5. Boundedness contradiction. From $\|\phi^{m}\|_\infty\le\|\phi\|_\infty\eta^{m-1}$, one gets both bounds in (26). The overlap term in (18) is bounded by $\|\Psi\|_\infty\|\Psi\|_1$, giving (27). Fubini transfers the essential bound to almost every spatial pair after time integration. Intersecting its full-measure set with the target identity inside $0<D<d_$ yields a positive-measure contradiction, so the argument is not merely pointwise at the diagonal.
6. Almost-everywhere and domain check. A pair-null set pulls back to a null set in triple or quadruple space. A triangle-null set can be used for a face only on

$\mathcal E_A=\mathcal D_4\cap\{x_A\in\mathcal D_3\}$. V02 makes precisely this restriction and does not impose the triangle-altitude condition on every tetrahedral face. Formula (33), which drops nonnegative third- and fourth-cumulant terms, remains valid throughout $\mathcal D_4$ without that substitution.

7. Scope of evidence. No numerical or empirical approximation is used in the theorem. The structural source claims are consistent with the frozen literature dossier, but the obstruction and repair arguments are self-contained enough that no unavailable source theorem carries a decisive proof step.

Decision record

Decision	Outcome	Basis
Does v02 establish the governing exact target at arbitrary prescribed intensity?	No	It supplies no exact witness and no general non-existence theorem.
Does v02 establish a correct restricted result?	Yes	The bounded pair-correlation argument excludes the stated $L^1\cap L^\infty$ scalar subcritical Hawkes subclass.
Is prescribed intensity handled within the restricted construction?	Yes	Equation (8) chooses the immigrant rate so that the finite-time projection has intensity exactly $\lambda _0$.
Are local fourth moments and absolute continuity through order four justified?	Yes	Equations (1)--(7), the measure induction after (16), (16a), and (22).
Are full product densities, rather than connected densities alone, used?	Yes	Equations (19)--(25) include every set partition through order four.
Are all prior author-facing issues closed?	Yes, as revision issues	Each requested repair is present and survives rechecking. The broad existence question remains unresolved.
Is another author repair round required?	No	No new mathematical or scope defect was found in the revised restricted theorem.

11. Critical review 4 — selected construction

Source: runs/20261019T114252Z_cox_rerun/reviews/approach_4/review_v02.md

The verdict rests on the following auditable checks.

1. Quantifier check. The submission consistently quantifies over every prescribed λ_0 only in the interval $(0,\lambda_*]$. It never promotes this to all positive intensities. The geometric domains match the frozen target, and the auxiliary enlargement of S_3 does not shrink a required domain.
2. Zero-cutoff check. Direct radial integration gives finite anchored mass for $1/D$ at the origin. The K_3 and K_4 marginal estimates remain finite and dominate the marginal inversion with no positive pair cutoff.
3. Positivity check. Each coefficient in E_2,E_3,E_4 was recomputed from the stated geometric bounds and partition counts. The displayed λ_* makes every residual kernel and every exact-size kernel nonnegative.
4. Intensity check. The relations among Z_r and Q_r, the parent rates $Z_r/r!$, and the singleton correction were recombined. All Q_3,Q_4 terms cancel and the remaining intensity equals λ_0 exactly.
5. Anchor and assignment check. The reanchoring map is an invertible integer block map of absolute determinant one. Symmetry and translation invariance then justify equality of injection contributions, giving the coefficient $1/(j-r)!$ rather than an assumed combinatorial factor.
6. Full-density check. All set partitions at orders two through four are present. The recursively defined q_r cancel precisely the lower partitions on the nested supports.
7. Process-property check. Bounded cluster size and range imply finite local exponential moments; diffuse parent positions and absolutely continuous nondegenerate relative marks imply simplicity and factorial-density absolute continuity.
8. Obstruction check. The global argument uses only the total mass of a finite reduced same-parent pair measure. Absolute continuity is introduced only on the target ball, where it follows from the assumed full pair identity after subtracting the Poisson baseline.
9. Source and provenance check. The frozen literature dossier supports the general Poisson-cluster partition calculus and identifies the local finite-cluster manuscript as exact only on its earlier restricted domain. V02 accurately labels its small-intensity statement as a reformulation and zero-cutoff re-derivation within that unpublished framework, not as an independent literature theorem or a new arbitrary-intensity mechanism.

Decision record

- Exact governing target at every prescribed positive intensity: not established; unresolved above the explicit threshold.
- Restricted exact theorem for every prescribed $0<\lambda_0\leq\lambda_*$: established by the checked argument.
- Full rather than connected densities: established; all parent partitions through order four are included.
- No-lower-cutoff pair domain: established; the $1/D$ singularity is locally integrable and is included for every positive distance up to M.
- Bounded-cluster obstruction: established only for the explicitly defined Poisson-cluster class with fixed finite K.
- All assigned v01 issues closed: yes.
- New actionable issues: none.
- Another scoped repair round: no. Further work would be new research on the arbitrary-intensity gap, not repair of this submission.

12. Critical review 5 — necessary conditions

Source: runs/20261019T114252Z_cox_rerun/reviews/approach_5/review_v02.md

The following are the decisive audit checks. They are a structured account of the verification, not a transcript of private deliberation.

Check	Calculation or evidence	Result
Governing quantifier	Claim section, final reasoning summary, limitations 3--4, and response item 1 all restrict existence to 0<\lambda_0\leq\lambda_*.	The arbitrary-intensity target is not overstated.
Face-support closure	For a face vertex, its opposite face-edge line lies in that vertex's tetrahedral opposite-face plane; distance to the line is no smaller than distance to the plane.	Every S_4 face lies in S_3; equation (11) only uses defined internal triangle values.
Nonempty tetrahedral support	The centroid identity bounds the least altitude by \sqrt{2/3}M; positive-measure nonemptiness gives strict inequality. A regular tetrahedron with a nearby admissible side length supplies an open positive set.	G>0, and Z_4,Z_3,Z_2 are strictly positive.
Endpoint integrability	4\pi\int_0^M r\,dr=2\pi M^2.	Removing the positive pair cutoff does not destroy local integrability or normalisation.
First marginal coefficient	With pair length d, A=dr/2; integrating 1/r over a radius-M ball gives \pi^2M^2.	\mathcal MK_3\leq2\pi^2M^2K_2=BK_2.
Second marginal coefficient	With signed height u, V=A|u|/3; integrating 3/(A|u|) over both ball cross-sections for H\leq|u|\leq M gives G/A.	\mathcal MK_4\leq GK_3.
q_2 positivity	The relative subtraction is at most \rho^2M/a_2=IMs=E_2(s).	The first bound in (15) follows when E_2\leq1/2.
q_3 positivity	Using A/D_{ij}\leq M/2 gives the pair contribution 6BMs; using A\leq M^2/2 gives 2BIM^2s^2.	Their sum is exactly E_3(s).
q_4 positivity	The four triple terms, three pair-pair terms, six pair terms and baseline term are bounded respectively by (16/3)GMs, (8BGM/I)s, 16BGM^2s^2, and (8/3)BGIM^3s^3.	Their sum is exactly E_4(s); no partition type is omitted.
Exact-size kernel positivity	Use the upper marginal bounds on q_4,q_3 together with a_4G=a_3/4 and a_3B=a_2/4.	The lower bounds (16) follow. The alternative expression for F_2 also gives F_2\leq q_2.
Prescribed intensity	From Q_2=a_2I-\rho^2L=\rho-\rho^2L, Z_4=Q_4, Z_3=Q_3-Q_4, and Z_2=Q_2-Q_3+Q_4/2, the non-singleton intensity plus (17) equals Q_2+\rho^2L=\rho.	The input intensity is obtained exactly throughout the stated interval.
Full-density hierarchy	Direct enumeration gives the all-singleton, pair-singleton, pair-pair, triple-singleton and same-parent blocks in counts 2,5,15.	Equations (19)--(21) are full factorial/product densities, not cumulants alone.
Process properties	Parent rates and cluster normalisers are finite; offsets are bounded; cluster size is at most four; all mark laws are absolutely continuous.	Stationarity, isotropy, simplicity, local finiteness, local exponential moments and density existence follow.
Null-window repair	Equation (28) now quantifies only over bounded positive-measure Borel windows and establishes 0<J(B_0)<\infty before division.	No undefined 0/0 case remains.
Count bound	Expanding (N-n)(N-n-1)\geq0 yields \mathbb E(N)_2\geq n(n-1)+2n\theta.	Equations (27)--(29), including the zero bound for mean below one, are correct.
Covariance bound	The variance identity includes \lambda_0\int f^2; diameter of the support ensures that every off-diagonal pair used lies in \mathcal D_2.	Equation (30) is a general point-process condition and does not import a Cox-only Jensen inequality.

Decision record

- Reviewed artefact: `tracks/approach_5/submissions/v02.md`, SHA-256 `3aa480e3279781cc0f45a009d71326c447778b94f626a7d88402f278347d3ab9`.
- Restricted-theorem decision: accepted as a correct self-contained informal proof at every prescribed 0<\lambda_0\leq\lambda_*(M,h_3,h_4), under the frozen positive-measure admissibility assumptions.
- Governing-target decision: not established; arbitrary prescribed positive intensity remains unresolved.
- Necessary-condition decision: equations (28)--(30) are valid but non-obstructive because C_2 is free.
- Prior-issue decision: A5-v01-01, A5-v01-07, A5-v01-08 and A5-v01-09 are all closed.
- Further repair decision: no additional author repair round is needed for this submission. Any continuation should be new research on the intensity gap, not another wording or local-proof repair.
- Permitted synthesis statement: approach 5 supplies an exact continuum construction with no lower pair cutoff for a nonempty interval of independently prescribed low intensities, together with general second-order necessary inequalities; it does not solve or refute the problem for all prescribed intensities.

13. Final synthesis — reasoning summary

Source: runs/20261019T114252Z_cox_rerun/review_report.md

The synthesis rests on the following auditable tests.

1. Quantifier test. Every claimed theorem was compared with the order `for every admissible geometry, for every prescribed lambda_0>0`. Only a closed interval starting at zero is proved. This alone prevents a favourable governing-target verdict.
2. Domain test. The pair law includes every `0<D<=M`; the triangle and tetrahedron domains use absolute altitude cutoffs; no triangle threshold is imposed on tetrahedral faces. The positive construction uses a larger internal triangle support rather than shrinking the target.
3. Full-density test. The positive construction, Cox carrier expansion and Hawkes hierarchy each include all `2,5,15` set partitions. Same-parent kernels and factorial cumulants are never substituted directly for `rho^(n)`.
4. No-cutoff test. `1/D` is locally integrable in three dimensions, and the exact marginal estimates remain proportional to the lower-order reciprocal kernels. This validates removal of the earlier `delta>0` cutoff.
5. Kernel-positivity test. Every lower partition at orders two through four is included in the residuals. The explicit error bounds imply nonnegative same-parent kernels, and the marginal inversion gives nonnegative exact-size cluster densities.
6. Process-law test. Parent rates, mark normalisation, reanchoring, singleton correction, local exponential moments, simplicity and factorial-density existence were checked before accepting the restricted theorem.
7. Intensity test. The finite-cluster intensity equals the prescribed input throughout the certified interval. Above the interval there is neither a construction nor a general obstruction. Cox mass scaling would solve intensity only after the missing exact directing measure is constructed.
8. Subclass-boundary test. Every negative result was traced to the hypothesis that makes it work: fixed cluster-size cap, Poisson uniform spheres, finite pointwise second moment, Hermitian determinantal structure, or essentially bounded Hawkes kernel. None was generalised beyond that class.
9. Excluded-shape test. Moment and covariance arguments were not allowed to fill low-altitude triangles, flat tetrahedra or separations beyond `M` with reciprocal formulas. This prevents spurious non-existence proofs.
10. Prior-work test. The spherical manuscript is approximate at prescribed intensity; the earlier finite-cluster manuscript is exact on a smaller domain and low intensity. Neither title or old audit verdict was treated as a solution of the frozen target.
11. Evidence test. No simulation, empirical catalogue scatter or finite grid was allowed to replace an exact continuum proof. Literature non-discovery was kept distinct from novelty or openness.

This is a structured account of the decisive checks and dependencies, not raw private chain-of-thought or a formal proof certificate.

Decision record

Decision	Final outcome	Basis
Does the rerun establish the exact target for every prescribed `lambda_0>0`?	No	All exact constructions stop at a finite certified low-intensity endpoint.
Does it establish exact existence for any nonempty intensity range?	Yes	The reviewed finite-cluster theorem covers every prescribed intensity in a closed nonempty interval starting at zero, with no pair cutoff and full `g_2,g_3,g_4`.
Is the restricted result merely approximate or numerical?	No	It is an exact analytic continuum construction with nonnegative cluster laws and complete partition cancellation.
Is general non-existence proved above the endpoint?	No	All obstructions are class-specific or constrain free amplitudes only.
Can the current size-at-most-four Poisson-cluster architecture cover all intensities?	No	The fixed-`K` pair-mass obstruction gives a finite intensity ceiling at fixed `M`.
Is prescribed intensity automatic in the Cox class?	Conditionally yes	Deterministic scaling works once one exact positive-intensity directing measure exists; no such general witness is known here.
Do the prior manuscripts settle the target?	No	One is approximate at prescribed intensity; the other is exact on a cutoff domain and at low intensity.
Was any coder or simulation evidence used?	No	All five reviewers found no useful specific numerical check; no coder was dispatched.
Is another prose-repair round needed for any v02 submission?	No	All assigned v01 repair items are closed and no delta review found a new actionable defect.
What remains?	New mathematics, not wording repair	Extend exact existence to arbitrary intensity or prove a valid general obstruction on only the frozen domains.

14. Recovered token and runtime ledger

Recovered after the run from cumulative local rollout trace events. Cached input is a subset of input; reasoning output is a subset of output; total equals input plus output. Runtime rows overlap and must not be summed.

Agent / session	Runtime	Input	Cached input	Uncached input	Output	Reasoning output	Total	Standard API-equivalent cost
Parent chat supervision and setup	03:47:06	40,895,045	40,731,008	164,037	23,758	6,445	40,918,803	$17.42
Coordinator	03:36:43	24,180,539	23,891,712	288,827	65,809	8,141	24,246,348	$12.03
Literature	00:18:35	11,506,253	11,102,848	403,405	43,730	14,489	11,549,983	$6.93
Proof 1, Poisson cluster, v01 + repair	00:21:47	7,190,086	6,714,112	475,974	53,236	25,732	7,243,322	$5.65
Proof 2, Cox, v01 + repair	00:23:43	6,615,750	6,327,296	288,454	63,742	33,688	6,679,492	$4.96
Proof 3, Hawkes, v01 + repair	00:22:05	6,999,088	6,700,800	298,288	64,778	31,021	7,063,866	$5.17
Proof 4, agent-selected, attempts + repair	00:25:48	9,901,952	9,583,616	318,336	63,413	30,885	9,965,365	$6.38
Proof 5, necessary conditions, v01 + repair	00:20:41	5,357,915	5,105,024	252,891	62,141	28,514	5,420,056	$4.30
Reviewer 1 v01	00:07:27	976,300	900,736	75,564	19,118	9,206	995,418	$1.04
Reviewer 1 v02 delta	00:04:56	1,457,007	1,384,832	72,175	15,586	7,578	1,472,593	$1.15
Reviewer 2 v01	00:08:35	1,888,810	1,783,296	105,514	21,029	10,503	1,909,839	$1.56
Reviewer 2 v02 delta	00:04:48	771,662	700,928	70,734	15,082	8,170	786,744	$0.86
Reviewer 3 v01	00:07:22	1,401,334	1,296,128	105,206	20,143	10,414	1,421,477	$1.34
Reviewer 3 v02 delta	00:06:12	1,377,263	1,294,592	82,671	18,999	6,391	1,396,262	$1.23
Reviewer 4 v01	00:07:31	1,270,065	1,180,416	89,649	23,214	13,468	1,293,279	$1.30
Reviewer 4 v02 delta	00:05:18	887,457	827,520	59,937	16,973	7,949	904,430	$0.91
Reviewer 5 v01	00:08:51	2,395,333	2,293,632	101,701	24,854	13,807	2,420,187	$1.82
Reviewer 5 v02 delta	00:06:36	741,233	672,512	68,721	19,415	10,924	760,648	$0.93
Final synthesis reviewer	00:10:10	6,053,589	5,838,464	215,125	31,685	9,134	6,085,274	$3.83
All recorded model sessions	03:47:06 task-turn envelope	131,866,681	128,329,472	3,537,209	666,705	286,459	132,533,386	$78.81

All recorded sessions: 131,866,681 input tokens; 666,705 output tokens, including 286,459 reasoning tokens; 132,533,386 total tokens; 03:47:06 task-turn envelope; USD 78.81 standard API-equivalent model cost. This is not evidence of a separate Codex subscription charge.

15. Source inventory and SHA-256 provenance

Each compiled section is traceable to a saved project file. The digest below is computed from the exact file used to build this PDF.

Artifact	Project-relative path	SHA-256
Run manifest	runs/20261019T114252Z_cox_rerun/run_manifest.json	4bffd4544c7f41ab2673df9ed16e44a381cdb9b477708b0e8b858871fafe0484
Literature dossier	runs/20261019T114252Z_cox_rerun/literature/dossier.md	5d42ad11b6b55246e6a0eb098b3f09c1bf3751cd3add4a819f5e427017b659f5
Proof 1 v02	runs/20261019T114252Z_cox_rerun/tracks/approach_1/submissions/v02.md	8e0835cf889621515534b83810cb5d00a7494117721fcd8461723a6e0008f160
Proof 2 v02	runs/20261019T114252Z_cox_rerun/tracks/approach_2/submissions/v02.md	0044e84da7c60a7bb4d8989724aa7001be0b1b7dd1759b08eb8f73c9e77aa49e
Proof 3 v02	runs/20261019T114252Z_cox_rerun/tracks/approach_3/submissions/v02.md	eb653072f665e0bd1fbb711d6d26297f9611a4673ad68c80922741a3f915dfa8
Proof 4 v02	runs/20261019T114252Z_cox_rerun/tracks/approach_4/submissions/v02.md	bdf6d669774a3155ac46a7ded6b978bd9246a7e074be45558c4be167b6848c2c
Proof 5 v02	runs/20261019T114252Z_cox_rerun/tracks/approach_5/submissions/v02.md	3aa480e3279781cc0f45a009d71326c447778b94f626a7d88402f278347d3ab9
Review 1 v02	runs/20261019T114252Z_cox_rerun/reviews/approach_1/review_v02.md	79857bc868c0ef121ea9dc570a17e77681a81de10388a93a4e36a000e9cb6703
Review 2 v02	runs/20261019T114252Z_cox_rerun/reviews/approach_2/review_v02.md	fd84230450c76fdc5232c576aa9747ee36c9a9edac2d48f3c244508ed9f3f077
Review 3 v02	runs/20261019T114252Z_cox_rerun/reviews/approach_3/review_v02.md	2921a482d9984451e7476645f5c8ef49b66aecb0f59635c75717974a4ded0c06
Review 4 v02	runs/20261019T114252Z_cox_rerun/reviews/approach_4/review_v02.md	4b2c848d6902c54368a8b0e62d38c07ad1eb13d7c7e6bd8983caf407c4a90d98
Review 5 v02	runs/20261019T114252Z_cox_rerun/reviews/approach_5/review_v02.md	b16c39a2260328a05fbf3f4b54220309bc54ba2a3e5a7955915328988dbfb1b7
Final synthesis	runs/20261019T114252Z_cox_rerun/review_report.md	61272d17668e44987293a7e17625247190a9339190e9fca782487448669c4140
Recovered accounting	runs/20261019T114252Z_cox_rerun/token_accounting_recovered.md	2309f7ec61b3c3c4940dc3d00bee5535e275f8de9c93c16cb3719861af3d4185

Bottom line. The workflow preserved substantial, auditable reasoning summaries and review checks. It did not, and cannot retroactively, preserve a textual transcript of hidden model deliberation. The reasoning-token ledger records quantity only.